

Kishan K

Software Engineer | AI and ML Engineer

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Portfolio: kishan-k.pages.dev

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GitHub: github.com/kishanaik5

LeetCode: leetcode.com/u/ZXO2yYfahk/

PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML)

10/2025 – Present

Silo Fortune Pvt Ltd

Bengaluru

- Designed and deployed the **Kisaan Sampurna** multimodal pipeline (Gemini 2.5 Flash + CV disease detection) delivering real-time advisories in **12+ Indian languages** with **92%+ precision**; integrated NDVI/UVI satellite indices and 7-day weather APIs.
- Built **Real-Time Mandi Price API** serving **7,000+ agricultural markets** with nearest-mandi auto-resolution via Haversine formula in SQL, 12-language support, and dual-source data (PostgreSQL + AWS S3 Parquet).
- Engineered **AI Farm Intelligence System** converting satellite indices (NDVI, NRI, EVI, DSWI) into **432 condition fingerprints** and 15-section advisories in **13 languages** via two-level LLM cache; deployed on **AWS ECS**.
- Built **AI Crop Stage Prediction** pipeline reading 5 years of satellite imagery to map **80+ sub-stages** for **21 crops** via parallel async Gemini calls, delivered in 10 languages via cache-first translation.

AI and Automation Intern

02/2025 – 08/2025

Sumedha Design Systems Pvt. Ltd

Hyderabad

- Engineered a scalable data extraction pipeline using **GCP Vision API** and **AWS S3** to process **10,000+** VLSI documents; orchestrated RAG-based systems for automated MCQ generation, reducing manual creation time.
- Architected an advanced retrieval framework leveraging **LlamaIndex** and **FAISS**, implementing hybrid search and re-ranking strategies that improved learner engagement by **15%** and reduced latency by **50%**.

TECHNICAL SKILLS

Programming: Python (AsyncIO), C/C++, SQL, JavaScript

AI & GenAI: Google Gemini, OpenAI, TensorFlow, PyTorch, Transformers, ViT, CNNs, RAG, NLP, AI Agents

Frameworks/Web: FastAPI, Django, Flask, Pydantic, SQLAlchemy, Alembic, React.js, React Native, Node.js

MLOps & Cloud: Docker, GitHub Actions, AWS (S3, RDS, SageMaker, ECS Fargate), GCP, Jenkins, Vercel

AI Frameworks/DB: LangChain, LlamaIndex, Rasa, Streamlit, PostgreSQL, MongoDB, FAISS, Redis

KEY PROJECTS

for more projects: kishan-k.pages.dev/#projects

Real-Time Mandi Price API

github.com/kishanaik5/mandi_price_forecast_predictor

FastAPI backend serving price data for 7,000+ agricultural markets with Haversine SQL nearest-mandi resolution, 12-language support, and dual-source data (PostgreSQL hot queries + AWS S3 Parquet for historical). Cascade dropdowns respond in under 5ms via in-memory cache.

- Tech Skills:** FastAPI, PostgreSQL, AWS S3, Docker, SQLAlchemy, JWT

AI Crop Stage Prediction

github.com/kishanaik5/Crop_stages_prediction_model

5-layer pipeline reading 5 years of satellite imagery (NDVI, EVI, NDMI) via VegetationIntelligenceEngine to detect growth cycles for 21 crops across 80+ sub-stages; generates pest/disease advisories via 16 parallel async Gemini calls in 10 Indian languages using cache-first translation.

- Tech Skills:** FastAPI, Gemini 2.0 Flash, AgroMonitoring API, PostgreSQL, AWS S3

AI Farm Intelligence System

github.com/kishanaik5/Yield-Weather-Forecast

Satellite indices (NDVI, NRI, EVI, DSWI) converted via deterministic bucketing into 432 condition fingerprints, then into 15-section LLM advisories in 13 Indian languages. Two-level Knowledge Base cache (English template + translation) drives LLM cost to zero (5,616 max cache entries). Deployed on AWS ECS Fargate + Jenkins CI/CD.

- Tech Skills:** FastAPI, Google Gemini, AWS S3, AWS ECS Fargate, PostgreSQL (JSONB), Jenkins

AI-Driven Sperm Mobility Analysis

tinyurl.com/sperm-mobility-project

Containerized FastAPI + Celery service for asynchronous semen video analysis; CV pipeline using YOLOv8 detection and DeepSORT tracking to compute sperm motility metrics, optimized for low-resource environments.

- Tech Skills:** FastAPI, Celery, YOLOv8, DeepSORT, AWS S3

EDUCATION

B.E. in Artificial Intelligence and Machine Learning

12/2021 – 07/2025

Ramaiah Institute of Technology, Bengaluru

CGPA: 8.6 / 10

PUBLICATIONS

Performance Evaluation of DL Models for Predicting Alzheimer's Disease,
Enhancing Predictive Maintenance with SHAP and LIME,

IEEE Bangalore, 2024
ICAI – ARSSS, 2025